

Preliminary Research Agenda (Rough Outline of Gameplan)

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SECTION 1: QUESTION

Refined Research Question:

How can latent space be operationalized as a designer-facing medium—linking rule-based parametric modeling with machine learning workflows—to support transparent and controllable design exploration?

Supporting Points:

Latent Space as a Medium:

- Investigate methods to extract and reveal core design dimensions from latent space—moving beyond merely reproducing geometric similarity.
- Aim to render latent representations intelligible so that designers can negotiate design outcomes.

Integration with Parametric Design:

- Establish a bridge between rule-based systems (e.g., Grasshopper) and ML-driven generative methods.
- Leverage latent disentanglement techniques to clarify abstract feature spaces.

Focus on Transparency and Control:

- “Transparent” means surfacing meaningful signals within the latent space, enabling informed design decisions.
- “Controllable” implies that designers can intentionally influence the generative process to align outputs with design intent.

SECTION 2: CONTEXT, PROBLEM STATEMENT, AND SCOPE

Context / Domain:

Domain & Background:

- Positioned at the intersection of computational design, machine learning, and HCI.
- Focused on design disciplines such as architecture, industrial design, and product design, where parametric tools (e.g., Grasshopper) already play a central role (Bolojan et al., Latent Design Spaces, 2024; Rhee & Veloso, Type Investigation, 2022).
- Integrates rule-based design with ML generative processes, reflecting established practices in the field.

Constraints:

Interactive Prototype:

- Develop an interactive system for early- to mid-stage design ideation rather than full-scale production.
- The scope will be limited to support preliminary user testing.

Emphasis on Transparency & Control

- Focus on yielding latent space representations that are interpretable and manipulable rather than producing merely aesthetic outputs (Zhou et al., Bridging Design Gaps, 2024).

Rationale:

Real-World Relevance:

- Grounding the research in environments where parametric design is standard ensures practical applicability and relevance.
- A constrained, interactive prototype enables iterative development and user-centered evaluation.

Addressing Current Gaps:

- Current ML systems often generate impressive yet opaque outputs (Dunnell & Lippman, Form Forge, n.d.).
- This research aims to reveal meaningful signals within latent space by applying latent disentanglement techniques, thus enhancing designer agency (Cardoso Llach, Sculpting Spaces, 2021; Cabezon et al., Feature Space Exploration, 2024).

Goals / Objectives / Purpose:

- Prototype Interface: Create a working prototype that enables designers to visually interact with and adjust latent space parameters.
- Workflow Integration: Develop design workflows that combine parametric modeling with ML-driven generative processes, fostering exploratory design.
- Theoretical Advancement: Contribute a conceptual model for understanding latent space as a medium of creative negotiation and controlled exploration.

SECTION 3: ANTICIPATED OUTCOMES

Prototype Interface:

- Develop a minimal yet effective interface that visualizes key latent dimensions and permits controlled adjustments.
- Allow designers to intuitively understand how variations in latent parameters impact design outcomes (Steinfeld, Significant Others, 2021).

Evaluation Framework:

- Establish metrics and user studies to evaluate usability, interpretability, and overall control offered by the prototype.
- Assess designer satisfaction, clarity of feedback, and the ability to iterate effectively using the tool.

Design Guidelines:

- Generate best practices for integrating ML-driven latent exploration into established parametric design workflows.
- Offer guidance on balancing automated generation with deliberate designer oversight.

Theoretical Contributions:

- Provide insights into how latent space can be reframed as a collaborative medium that enhances creativity.
- Offer evidence for the role of latent disentanglement in revealing actionable design parameters.

SECTION 4: SIGNIFICANCE OR IMPORTANCE

Bridging Design Paradigms: Creates a tangible link between rule-based parametric design and ML's generative capabilities, uniting two traditionally separate approaches.

Reconceptualizing Latent Space: Reframes latent space as an active, interpretable medium rather than an opaque computational backend.

Enhancing Designer Agency: Empowers designers by providing tools that reveal and control latent variations, supporting intentional and iterative creative decision-making.

Practical Industry Impact: Informs the development of ML-integrated tools that are both usable and aligned with creative design workflows.

Contribution to Critical Discourse: Adds to scholarly debates on ML interpretability and human-AI collaboration, building on insights from works such as Steinfeld (Significant Others, 2021) and Cardoso Llach (Sculpting Spaces, 2021).

SECTION 5: RELEVANCE TO YOU

Prior Industry Experience: Experience in architectural design and prototype development using advanced computational tools provides a practical perspective on integrating design methodologies.

Academic and Technical Background: Coursework in deep learning, machine learning, and generative AI has furnished both the theoretical and technical foundations necessary to critically engage with ML methodologies.

Alignment with Research Goals: The research addresses a clear need for tools that enhance interpretability and control—topics that are central to current debates in ML-assisted design.

Unique Positioning: Combines practical design process experience with a solid technical background to explore how latent space can serve as a collaborative, designer-facing medium.

Addressing Key Blind Spots: Recognizes current limitations in scaling ML training pipelines and deployment strategies, and focuses on interactive

demos and interpretive evaluations to advance the field.

PRELIMINARY REFERENCES

- [1] Bolojan, D., Yousif, S., & Vermisso, E. (2024). Latent Design Spaces: Interconnected Deep Deep Learning Models for Expanding the Architectural Search Space. https://doi.org/10.1007/978-3-031-36922-3_13
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- [4] Dunnell, K., & Lippman, A. (n.d.). Form Forge: Latent Space Exploration of Architectural Forms via Explicit Latent Variable Manipulation.
- [5] Rhee, J., & Veloso, P. (2022). Type Investigation in the Form of High-Rise Building Using Deep Neural Network.
- [6] Steinfeld, L. (2019). Significant Others: Machine Learning as Actor & Provocateur in Art and Design.
- [7] Zhou, R., Yuan, C., Permenter, F., Zhang, Y., Arechiga, N., Klenk, M., & Ahmed, F. (2024). Bridging Design Gaps: A Parametric Data Completion Approach With Graph Guided Diffusion Models.

